

Task 3:

1) Design a function `greet()` to print "Hello!" and call it from `main()`

```
#include <stdio.h>
```

```
void greet() {
```

```
    printf("Hello! \n");
```

```
}
```

```
int main() {
```

```
    greet();
```

```
    return 0;
```

```
}
```

output:

Hello !

2) write a function `add()` that takes two integers as parameters and prints their sum

```
#include <stdio.h>
```

```
void add (int a, int b) {
```

```
    printf("Sum: %d \n", a+b);
```

```
}
```

```
int main() {
```

```
    int x, y;
```

```
    printf("Enter two integers: ");
```

```
    scanf("%d %d", &x, &y);
```

```
    add(x, y);
```

```
    return 0;
```

```
}
```

output:

Enter two integers:

5 7

Sum: 12

3) Create a function multiply() that returns the product of two numbers

```
#include <stdio.h>
```

```
int multiply(int a, int b) {
```

```
    return a * b;
```

```
}
```

```
int main() {
```

```
    int x, y, result;
```

```
    printf("Enter two integers: ");
```

```
    scanf("%d %d", &x, &y);
```

```
    result = multiply(x, y);
```

```
    printf("product: %d\n", result);
```

```
    return 0;
```

```
}
```

output:

Enter two integers: 4 0

Product: 24

4) Implement a recursive function to calculate the factorial of a number

```
#include <stdio.h>
```

```
int factorial(int n) {
```

```
    if (n == 0 || n == 1)
```

```
        return 1;
```

```
    else
```

```
        return n * factorial(n-1);
```

```
}
```

```
int main() {
```

```
    int n;
```



```

printf("Enter a number:");
scanf("%d", &n);
printf("Factorial of %d is %d\n", n,
factorial(n));
return 0;
}

```

output:

Enter a number: 5
Factorial of 5 is 120

5) Program with two function: one to square a number and another to display it.

```

#include <stdio.h>

```

```

int square(int n) {
    return n * n;
}

```

```

void display(int n) {
    printf("Result: %d\n", n);
}

```

```

int main() {

```

```

    int num;

```

```

    printf("Enter a number to square:");
    scanf("%d", &num);

```

```

    int sq = square(num);

```

```

    display(sq);

```

```

    return 0;
}

```

output:

Enter a number to
square: 9
Result: 81